

CoMem: Persistent Intermediate-Residual Memory for Bounded-Read Long-Context Inference

Anonymous ACL submission

Abstract

Reusable chunk caches precompute document state, but the amount of model depth that should be prepaid remains unclear. We study this question with **CoMem**, which stores one depth- j residual per context token, retrieves a bounded chunk set, and resumes only layers $[j:L)$. The main result is a matched depth-reuse frontier: with identical selected chunks, order, mask, examples, and adapter, $j=12$ reduces isolated model Read from 931.9 to 664.4 ms ($1.403\times$) while lowering a 15-cell RULER macro from 99.19 to 96.07. A rank-32 self-distillation adapter makes this operating point usable without updating the backbone. The one-time Write amortizes only under reuse: measured break-even is about 8–11 queries at 32k and 26–28 at 128k, depending on generation length and store placement. We also report a transparent negative result: at matched online latency, raw-text replay scores 64.78 versus 53.22 for CoMem on a held-out mixed diagnostic cohort. Finally, a paired multikey factorization bounds one failure mode. Restoring lower-layer document context raises 92.5 to 100.0, whereas changing positions alone lowers accuracy; a deployable 32-token overlap recovers to 98.5 without increasing persistent bytes or per-query Read work. These findings establish a measured quality–latency–storage frontier and a narrow mechanism diagnosis, not superiority over raw-text retrieval or existing modular-cache systems.

1 Introduction

Repeated-query long-context serving can save work in two different ways. Raw-text retrieval bounds the online token set but recomputes all layers (Lewis et al., 2020; Xu et al., 2024). Prefix, position-independent caching (PIC), and modular-cache systems instead reuse precomputed document state, usually layer-wise KV, while repairing position or context dependencies (Yao et al.,

2024; Lu et al., 2024; Hu et al., 2025; Agarwal et al., 2025; Chen et al., 2026). This paper asks a narrower question: *how much transformer depth should a reusable document object prepay?*

We instantiate that question with **CoMem** (Figure 1). WRITE runs each context chunk through layers $[0:j)$ and stores one residual vector per token. SELECT chooses a fixed number of chunks. READ packs their residuals with the query and executes only $[j:L)$ with full cross-pack attention. The endpoint $j=0$ stores token IDs and replays the same selected pack through every layer. Thus j varies prepaid depth while holding evidence fixed.

Our strongest evidence is this *matched depth-reuse frontier*. On Qwen3-8B, $j=12$ lowers isolated selected-pack Read from 931.9 to 664.4 ms ($1.403\times$) but lowers a paired 15-cell RULER macro from 99.19 to 96.07. This is not quality-preserving acceleration. A self-distilled upper-layer LoRA is necessary to make the residual interface usable, and the 8 KiB/token store is worthwhile only when reused: measured break-even ranges from 8–11 queries at 32k to 26–28 at 128k. Much larger dense-context numbers combine bounded selection with depth reuse and are secondary operating points, not the causal depth result.

We deliberately test the strongest counterclaim. At matched online latency, raw-text replay can spend its budget on ten chunks while CoMem reads twelve; it still wins 64.78 to 53.22 on a held-out mixed diagnostic cohort. CoMem therefore does not establish a new quality ranking over RAG or reusable-KV systems. It establishes an internally matched computation–quality–storage frontier.

We also bound one source of error rather than claiming a complete explanation. On a paired 8k/16k multikey diagnostic, document-contextual lower-layer states close the 92.5-to-100.0 gap, while document-origin position remapping alone

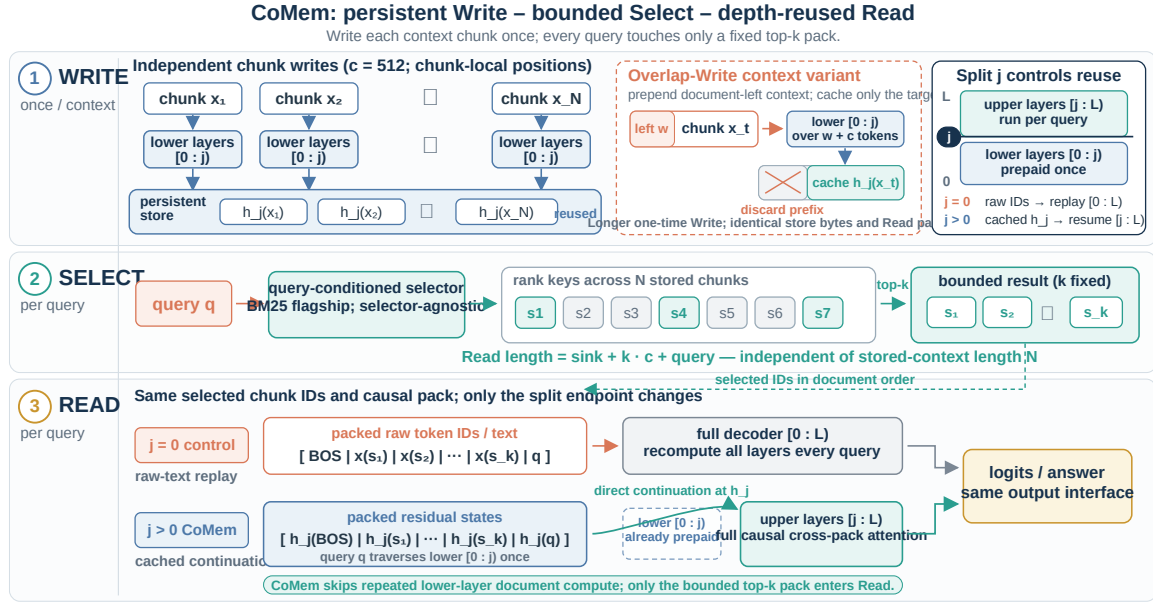


Figure 1: **CoMem treats transformer depth as a cross-query reuse axis.** WRITE stores one intermediate residual per token, SELECT retrieves a fixed working set, and READ resumes the upper layers with the query present. The matched $j=0$ path retrieves the same raw-text chunks but recomputes every layer. A small left-overlap is evaluated as a focused repair for missing lower-layer document context.

reduces accuracy to 88.0. A deployable 32-token left overlap reaches 98.5 with unchanged persistent bytes and per-query Read work. Because this repair is not evaluated on the full natural-task suite, we treat it as a localized diagnosis and engineering hypothesis.

Contributions.

- **Matched frontier:** a same-evidence, same-adaptor measurement of the incremental benefit and quality cost of reusing lower-layer computation.
- **Transparent system boundary:** persistent bytes, Write amortization, serving crossover, and an equal-latency negative result, with bounded selection kept separate from depth reuse.
- **Bounded diagnosis:** controlled continuous-state, context, position, and overlap variants that isolate a major Write-side failure on one paired multikey cohort without generalizing it beyond the evidence.

2 Related Work

We position CoMem by stored object and post-retrieval computation. Table 1 is an interface map, not an empirical ranking.

PIC and modular document caches. CacheBlend, TurboRAG, RAGCache, and Cache-Craft precompute or reuse retrieved chunk KV and handle context-dependent composition in different ways (Yao et al., 2024; Lu et al., 2024; Jin et al., 2024; Agarwal et al., 2025). PIC systems such as EPIC and MEPIC splice independently encoded per-layer KV and repair boundaries or selected tokens; APE uses parallel context encoding and attention realignment (Hu et al., 2025; Wang et al., 2025; Yang et al., 2025b). Fusion RAG Cache studies selection and cache fusion at serving scale (Wang et al., 2026). Learned modular objects further include KV Packet’s independently compiled KV plus boundary adapters and Cartridges’ distilled reusable KV representations (Chen et al., 2026; Eyuboglu et al., 2025). Concurrent Cartridges at Scale and SemPIC retrieve learned per-document KV objects (Hardalov et al., 2026; Xie et al., 2026). These works already establish reusable document caches. CoMem does *not* claim that idea; it studies the distinct single-residual object, whose split depth trades persistent bytes and online upper-layer computation. We lack a matched same-backbone implementation of these systems, so Table 1 supports taxonomy rather than superiority.

Family	Persistent object	Composition / repair	Post-selection model work	Relation to CoMem
Text RAG (Lewis et al., 2020)	token IDs	concatenate retrieved text	all layers	matched $j=0$ endpoint
Chunk-KV reuse (Yao et al., 2024; Lu et al., 2024; Agarwal et al., 2025)	per-layer KV	fuse / partially recompute	repair or decode	closest serving workload
PIC / parallel encoding (Hu et al., 2025; Wang et al., 2025; Yang et al., 2025b)	per-layer KV	position/boundary repair	selected repair or decode	independent modular chunks
Learned modular KV (Chen et al., 2026; Eyuboglu et al., 2025; Xie et al., 2026)	learned per-layer KV	trained compiler/adapters	little/no document recompute	closest learned objects
Activation replay (Gao et al., 2025; Qasim et al., 2026)	activation or residual	restore/reconstruct state	replay suffix or KV	intermediate-state precedent
CoMem	one depth- j residual/token	pack selected residuals	execute layers $[j:L)$	explicit depth-reuse frontier

Table 1: **Positioning among reusable-context interfaces.** This taxonomy does not rank systems: the cited methods differ in backbone, training, workload, hardware, storage tier, and timing boundary. CoMem’s specific object is one residual per token at a tunable depth; the empirical claim in this paper is the matched $j=0$ versus $j=12$ frontier, not superiority over these families.

Intermediate-state reuse and token reduction.

ReadOnce Transformers and Embedding Recycling cache intermediate text representations and adapt later layers (Lin et al., 2021; Saad-Falcon et al., 2023). HCache checkpoints activations to restore evicted state; KV-Direct reconstructs layer-wise KV from residuals; ILRe and REFORM select or gather tokens before recomputation (Gao et al., 2025; Qasim et al., 2026; Liang et al., 2025; Song et al., 2025). GemFilter likewise selects with early layers but does not persist the states across queries (Shi et al., 2024). Token/KV compression instead changes the retained token set or state budget after full or partial processing (Xiao et al., 2024b; Zhang et al., 2023; Li et al., 2024; Cai et al., 2024; Liu et al., 2024). CoMem’s narrow technical object is one persistent depth- j residual per selected token followed by direct execution of $[j:L)$.

External memory and the matched endpoint.

MemoryLLM, LongMem, and XC-Cache add latent pools or auxiliary readers (Wang et al., 2024, 2023; Monteiro et al., 2024); they are useful external references but not matched depth controls. Our $j=0$ endpoint is deliberately narrower than “RAG”: it fixes selector, chunk IDs and order, sink, mask, examples, and adapter, then replaces stored residuals with token IDs. The paper’s causal claim is therefore about depth reuse under identical evidence, not a ranking over retrieval or memory systems.

3 Why a Depth Frontier?

The interface does not require a layer at which a model has “understood” the document. It requires only that intermediate states retain useful information and that later layers can be adapted to read states produced without the query. Figure 2 supplies motivation, while the matched experiment in Section 5 supplies the actual evidence.

A frozen interface deteriorates as the split moves deeper. With separately distilled adapters,

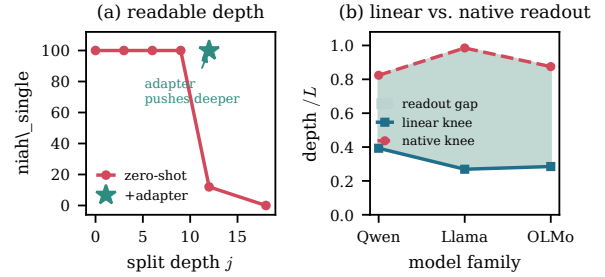


Figure 2: **Intermediate accessibility motivates, but does not validate, depth reuse.** Frozen probes become informative before native LM-head readout, but both measures are protocol-dependent; definitions and controls are in Appendix A.1.

RULER changes from 98.29 at $j=6$ to 96.07 at $j=12$ and 55.41 at $j=18$, while fixed-pack Read speedup rises from $1.166\times$ to $1.403\times$ and $1.807\times$ (Appendix Table 9). Because each split has a different adapter span and parameter count, this is a deployment curve rather than a compute-matched causal ablation.

Selection and representation also fail separately. BM25 retains lexical single-needle evidence, but on LongEval the distilled $j=12$ reader reaches only 64–76% accuracy conditional on a retrieval hit; BABILong exhibits both misses and hit-conditional readout failures. We therefore evaluate depth under a fixed evidence pack and diagnose Write context only after retrieval is held constant.

4 CoMem

Interface. Let a decoder have L layers and width d , with half-open layer ranges and h_j denoting the input to block j . WRITE divides a document into c -token chunks, runs each through $[0:j)$ with positions restarted at zero, and stores $h_j \in \mathbb{R}^{c \times d}$. SELECT returns top- k chunk IDs. READ orders them by document position, assigns fresh contiguous positions to $[\text{BOS}; \{h_j\}; h_j(q)]$, and runs $[j:L)$ with full causal cross-pack attention. For $j=0$, the same path stores token IDs and

executes all layers, giving the matched endpoint of the depth axis.

Storage and bounded model work. With head width d_{head} , query-head count n_q , and KV-head count n_{kv} , residual-to-full-KV bytes/token at a common dtype are

$$\frac{|h_j|}{|\text{KV}|} = \frac{d}{2Ln_{kv}d_{\text{head}}} = \frac{n_q}{2Ln_{kv}}, \quad (1)$$

using $d = n_q d_{\text{head}}$. For Qwen3-8B this is 1/18: 8 KiB rather than 144 KiB/token in bf16. It is about 1 GiB at 128k tokens—far larger than text, so reuse is essential. The model-side Read length is sink + kc + query (nominally 6,657 tokens for $k=12, c=512$), independent of stored-context length. The persistent store, index, and selector are not bounded.

Overlap-Write. Independent writes omit lower-layer left context. For the diagnostic variant, chunk i is prefixed by its preceding w tokens during Write, but only the current c residuals are stored. Thus persistent bytes, selected-pack length, and per-query Read/decode are unchanged; Write work and edit invalidation grow. This is a tested repair on one diagnostic, not the default for the full suite.

Self-distillation. Following knowledge distillation and low-rank adaptation (Hinton et al., 2015; Hu et al., 2022), a $j=12$ student matches a frozen $j=0$ teacher on the teacher’s top-64 logit support. For renormalized teacher and student distributions p and q , respectively,

$$\mathcal{L}_{\text{distill}} = 0.6 \text{KL}(p||q) + 0.4 \text{KL}(q||p), \quad (2)$$

with no cross-entropy term. The rank-32 LoRA ($\alpha=64$, zero dropout) updates attention and MLP projections in layers 12–35; the Qwen3-8B backbone stays frozen. Training uses 4,000 bf16 DDP steps on 2,048-token PG-19 windows (Rae et al., 2019), without benchmark supervision, retrieval, or synthetic long-context data. Full optimizer and artifact details are in the appendix.

Selection and generation. The flagship selector is deterministic token-ID BM25 (Robertson and Zaragoza, 2009), chosen to avoid a separately trained retriever. It uses top-12, hop width four, and three frontier-expansion rounds: round 1 scores the question; later rounds concatenate

tokens from newly selected chunks, score unselected chunks, and break ties by document index. Frozen dense retrieval is reported only as a diagnostic. During generation, lower layers cache the query/generation prefix and upper layers cache the selected pack; the retrieved pack is not rerun for each output token. Exact masks, positions, and timing boundaries appear in the appendix.

5 Experiments

Setup. The principal model is frozen Qwen3-8B (Yang et al., 2025a), with $j=12$, 512-token chunks, iterative BM25 top-12, a BOS sink, and the rank-32 LoRA. Prompting is plain text without chat templates. We evaluate RULER (Hsieh et al., 2024), BABILong (Kuratov et al., 2024), LongEval (Li et al., 2023), six LongBench QA datasets (Bai et al., 2023), and all 1,986 LoCoMo questions (Maharana et al., 2024). Appendices A.2 and B give complete cells, prompts, sample counts, uncertainty, and integrity checks. We keep four timing boundaries separate: selected-pack model Read, store-ready online prefill, Write-inclusive pipeline, and external index/store cost.

5.1 Matched depth reuse

Table 2 is the central comparison. Raw-text replay ($j=0$) and CoMem receive the same selected chunks in the same order and use the same sink, mask, examples, and LoRA. Skipping the lower 12 layers reduces fixed-pack Read from 931.9 to 664.4 ms (1.403 $\times$), but lowers the RULER macro from 99.19 to 96.07. CoMem is therefore not quality preserving. The continuous-prefix control exactly matches $j=0$, showing that upper-layer continuation can reproduce replay when supplied compatible states. Because it jointly restores document, cross-chunk, and query-side lower-layer interaction, the control is a fidelity ceiling rather than a single-factor explanation.

The same quality ordering largely holds outside RULER: matched $j=0$ scores 97.2 versus 69.0 on LongEval, 41.59 versus 38.27 on LoCoMo, and 12.31 versus 12.15 on six LongBench QA datasets; task-level BABILong results are mixed. We therefore use these tables as scope checks, not a heterogeneous leaderboard.

5.2 Amortization and the negative frontier

The residual store costs 8,192 bytes/token and must first be written. In the write-once serving

Operating point	Stored object	Replay	RULER↑	Read	Write	Store/token
Matched raw-text replay ($j=0$)	token IDs	36 layers	99.19	931.9 ms	on query	~4–8 B
CoMem + LoRA ($j=12$)	h_{12}	24 layers	96.07	664.4 ms	once	8,192 B
Continuous-prefix control	h_{12}	24 layers	99.19	—	non-deployable	8,192 B

Table 2: **The central result is a matched depth-reuse frontier, not a cross-system ranking.** Quality uses the paired 15-cell RULER-B cohort ($n=1,500$); the $j=0$ – $j=12$ gap is 3.12 points with paired-bootstrap 95% CI [2.36, 3.93]. Read latency uses the same LoRA and a fixed selected pack (mean approximately 6.5k tokens; nominal cap 6,657) from 16k source contexts (three process medians); retrieval, persistent I/O, reusable Write, and decode are excluded. The continuous-prefix control supplies layer-12 states from a single causal lower-layer forward and exactly matches raw-text replay, showing that upper-layer continuation can be faithful with compatible states. The control jointly restores document, cross-chunk, and query-side lower-layer interaction, so it is a fidelity ceiling rather than a single-factor attribution.

Store	$G=1$	$G=32$	$G=128$	$G=512$
32k, CPU-pinned	8.9	9.2	10.9	94.0
32k, GPU-resident	8.4	7.7	5.5	—
128k, CPU-pinned	27.6	—	—	—
128k, GPU-resident	25.8	—	—	—

Table 3: **Measured repeated-query break-even Q^* .** G is generated tokens; a cell is the number of queries after which cumulative CoMem time is lower than matched $j=0$ raw-text replay. Both arms share selection, example, and pack; selection cancels in $Q^* = (W_{\text{CoMem}} - I_{j=0}) / (T_{j=0} - T_{\text{CoMem}})$. The 32k CPU cell uses $W_{\text{CoMem}}=2.253$ s and a 2.16 ms raw-token index build. Values derive from three process medians and include store fetch, model Read, and decode; raw component times and dispersion are archived. “—” denotes an unavailable or non-finite retained cell.

harness, CoMem overtakes matched raw replay after 8–11 queries at 32k for generated lengths up to 128 tokens; the 512-token CPU case moves to 94 queries because decode dominates. At 128k and one generated token, break-even is 25.8/27.6 queries for GPU/CPU placement. Each value follows the displayed measured-time equation and includes fetch, model Read, and decode; for one-off or weakly reused documents, raw text is cheaper.

We then ask whether cheaper replay can buy enough extra evidence to improve quality at equal online latency. A latency-only calibration chooses raw-text top-10 versus CoMem top-12; on the held-out cohort, raw replay leads by 11.56 points. This is the paper’s decision-relevant negative result: saved lower-layer compute does not compensate for residual-interface loss in this setting.

5.3 Bounded diagnosis

Holding retrieval, adapter, and upper-layer Read fixed, the 2×2 diagnostic separates lower-layer attention scope from Read positions. Document-

Method	top- k	Matched quality↑	Online latency
Raw-text replay	10	64.78	within $\pm 5\%$
CoMem + LoRA	12	53.22	anchor

Table 4: **Equal-latency BM25 comparison is negative for CoMem.** The retrieval budget is chosen on a disjoint latency-only calibration split, then frozen and evaluated on a mixed diagnostic cohort. CoMem trails raw-text replay by 11.56 points (paired-bootstrap 95% CI [−14.44, −8.67]). Thus the extra evidence admitted by cheaper upper-layer replay does not compensate for residual readout loss in this cohort.

Lower-layer Write scope	local Read pos.	document-origin Read pos.
Chunk local	92.5	88.0
Document contextual	100.0	100.0

Table 5: **Bounded context-scope $\times$ Read-position diagnosis.** Paired Qwen3-8B multikey macro over 8k/16k ($n=200$), with identical retrieval, LoRA, and upper-layer Read. Document context adds 7.5 points with local Read positions and 12.0 with document-origin positions; changing position alone lowers the chunk-local writer by 4.5 points. The +4.5 interaction means the factors are not additive. This attribution is limited to the displayed cohort.

contextual Write closes the gap under either position convention; document-origin positions alone make chunk-local states 4.5 points worse. The +4.5 interaction precludes an additive attribution, but bounds a major tested failure to the Write representation rather than position remapping.

Overlap-Write turns this control into a local deployable intervention: $w=32$ raises 92.5 to 98.5 and $w=128$ to 99.0 while preserving stored bytes and per-query work. Its measured wall-clock overhead exceeds the theoretical FLOP ratios, and it is evaluated only on this paired synthetic cohort. We therefore claim a bounded diagnosis, not a universal repair.

Writer	Context	Macro $\uparrow$	Write FLOPs
Independent ($w=0$)	current chunk	92.5	1.000 $\times$
Overlap ($w=32$)	32-token left prefix	98.5	1.057 $\times$
Overlap ($w=64$)	64-token left prefix	98.5	1.115 $\times$
Overlap ($w=128$)	128-token left prefix	99.0	1.229 $\times$
Document-context control	full prefix	100.0	$O(N)$ forward

Table 6: **Lower-layer context is a focused repair, not a new headline frontier.** Paired Qwen3-8B results on RULER niah_multikey_1 at 8k and 16k ($n=200$ pooled). Every overlap width significantly exceeds the independent writer; for $w=32$, the gain is +6.0 points, 95% CI [3.0, 9.5]. Persistent storage and Read/decode are identical across deployable rows. FLOPs are theoretical lower-12 Write ratios; measured wall-clock overhead is larger because of chunk-level execution. No natural-task overlap result is claimed.

Secondary scope checks. A frozen BGE-large-en-v1.5 diagnostic shows that retrieval recall can instead be the bottleneck (Appendix A.1). Store-ready dense-prefill and Write-inclusive measurements, synthetic 128k stress tests, and cross-scale/MoE ports are reported in the appendix. They support bounded-read operation and implementation portability, but do not alter the matched frontier above.

6 Conclusion

CoMem makes transformer depth an explicit cross-query reuse axis: contexts are written to intermediate residuals, a bounded set is selected per query, and only the upper layers are resumed. The contribution is the measured frontier, not a claim of free compression or a leaderboard win. With identical evidence and adapter, $j=12$ yields 1.403 $\times$ faster model Read at a 3.12-point RULER cost, and its 8 KiB/token store pays off only after repeated queries. At equal online latency, raw-text replay remains substantially more accurate. A paired context–position factorization further localizes one synthetic multikey gap to the lower-layer Write representation, and overlap recovers most of that local gap without enlarging the store or Read work. The next decisive step is a same-backbone, end-to-end comparison with PIC and learned modular-KV systems, plus natural-task validation of contextual writing. Until then, CoMem should be read as a transparent depth-reuse design point with a bounded diagnosis.

Limitations

CoMem is evaluated primarily on one 8B backbone with a lexical selector and an English

benchmark suite. Results on other Qwen scales and sparse MoEs support portability but are not matched replications, and several large-model cells use smaller sample counts. The principal adapter is trained once; most ablations measure evaluation uncertainty rather than training-seed variance.

The paper does not provide a same-backbone, same-hardware implementation of the closest PIC, chunk-KV repair, or learned modular-cache systems. Their published operating points use different models, tasks, training budgets, storage tiers, and latency boundaries, so Table 1 is taxonomic and our experiments establish an internal depth frontier rather than competitive superiority.

The system assumes repeated queries over a relatively stable corpus. Its residual store grows linearly and is much larger than raw text or token IDs; for one-off queries, frequently edited documents, or low reuse, raw-text retrieval is usually preferable. Selector and external-store latency still grow with corpus size even though model-side Read is bounded. Reported crossover and speed measurements depend on hardware, kernels, batch size, storage tier, and the precise timing boundary.

Stored states are tied to the exact backbone, tokenizer, split depth, adapter, and lower-layer implementation; model upgrades generally require rewriting the store. Overlap-Write also weakens strict chunk independence because a local edit can invalidate the following chunk’s cached state. We do not evaluate residual quantization, eviction, or multi-tenant update contention.

Quality is task dependent. Bounded top- k retrieval imposes an evidence ceiling, lexical selection transfers poorly to low-overlap or non-whitespace languages, and residual readout can fail even when the evidence is present. Overlap-Write is tested on a focused multikey diagnostic rather than the natural benchmark suite, and its wall-clock Write overhead is not integrated into a repaired end-to-end frontier. It is not a universal quality fix. Results beyond the backbone’s native context window are positional-extrapolation stress tests.

Our probes and controls characterize readout compatibility; they do not locate “understanding,” factual storage, or causal necessity at a particular layer. Long-form generation, code, multilingual and multimodal memory, dynamic updates/deletions, stronger learned retrievers, and production-scale multi-tenant scheduling remain

open. The serving experiments report medians on single-query harnesses rather than concurrent p95 throughput or tail latency.

7 Ethical Considerations

CoMem changes the cost and storage organization of an underlying language model; it does not add a safety mechanism. It therefore inherits risks such as hallucination, biased generation, unsafe completion, and unintended disclosure of sensitive text selected from a memory store. Lower-cost access to long histories could also scale benign applications and deliberate misuse alike. Cached residual tensors may also expose source information through inversion or membership-inference attacks; they should receive protections at least as strong as the source text. We do not empirically evaluate these attacks, so the risk is a deployment concern rather than a demonstrated vulnerability. Their large persistent volume and possible cross-tenant reuse make authorization, isolation, and deletion operational requirements rather than properties conferred by non-readable tensor storage. Deployments should encrypt and audit the persistent store, enforce access control, filter or redact sensitive sources before Write, support verifiable deletion, restrict retrieval to authorized material, and retain application-appropriate output monitoring and human review. Although bounded reads reduce accelerator memory and per-query computation, training and large-scale evaluation still consume energy; we report the measured hardware and final training budget in Appendix A.4.

We collect no new human-subject data and recruit no annotators. Our experiments use previously released model, corpus, and benchmark artifacts; LoCoMo’s released conversations are generated by paired language-model agents rather than newly collected private conversations. The anonymous release excludes model weights and benchmark text. Subject to benchmark and provider terms, it includes the adapter, evaluation scripts, prediction hashes or permissible shards, judge prompts/parsed decisions, and raw timing records; credentials and private API metadata are excluded.

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Length	Dense prefill	CoMem prefill	Speedup	Dense peak	CoMem peak
8k	1.20 s	1.05 s	1.14×	18.8 GB	18.7 GB
32k	7.33 s	1.06 s	6.92×	25.0 GB	18.7 GB
128k	71.37 s	1.10 s	64.9×	50.0 GB	18.7 GB

Table 7: Select-first online prefill on one H20 (Qwen3-8B, bf16/SDPA, median of three after one warmup). Dense runs the stock full-context path; CoMem uses its rank-32 LoRA and a fixed 6,657-token top-12 pack. This cohort times online model prefill after the persistent memory is built; it excludes one-time document Write and external-store fetch. The speedup is therefore an online bounded-read operating point, not a production end-to-end or depth-only causal effect.

A Additional Results and Ablations

This appendix contains the detailed task tables, mechanism ablations, and cross-scale generality results summarized in the eight-page main paper. Appendix B separately documents aggregation, uncertainty, and the LoCoMo judge protocol.

Secondary dense-context operating point.

When the persistent store already exists, the select-first path reads a fixed 6,657-token pack at every source length. Table 7 reports 1.10 s and 18.7 GB at 128k versus 71.37 s and 50.0 GB for dense prefill on the same H20. The resulting 64.9× combines bounded selection with depth reuse; it is neither the incremental depth effect nor an end-to-end repeated-query comparison.

Content-depth probe protocol. For SST2, WiC, and RTE we extract every layer once from a fixed stratified pool (up to 3,000 train examples), then fit L2 logistic regressions on five independent 60/20/20 train/dev/test splits. Features are standardized; $C \in \{0.1, 1, 10\}$ is selected on development data. For layer accuracy $a(l)$, the knee is $\min\{l : a(l) \geq 0.98 \max_l a(l)\}$. The native readout instead applies the model’s own final norm and LM head to each layer using fixed verbalizers. Lexical uni/bi-grams, token length, random labels, majority class, and balanced training serve as controls. On SST2, probe peak is about 0.90 versus 0.71 lexical, 0.56 position/majority, and 0.54 random-label accuracy; WiC/RTE are lower-selectivity and have wider per-task intervals. The reported confidence intervals are Student- t intervals over task-split knees, not training-seed uncertainty.

Configuration	Split j	LoRA	LoCoMo Judge	BABILong mean
				qa1 / qa2 / qa5
Matched raw-text replay	0	no	41.59	66.3 / 38.9 / 63.3
CoMem frozen	6	no	32.78	44.6 / 22.6 / 53.9
CoMem frozen	9	no	29.15	42.4 / 19.6 / 55.6
CoMem frozen	12	no	24.52	33.4 / 18.0 / 60.3
CoMem distilled	12	yes	38.27	55.6 / 27.0 / 68.7

Table 8: Controlled comparison of retrieval, split depth, and self-distillation using Qwen3-8B. All configurations follow the same plain-text prompting protocol, with model-specific chat templating disabled (`chat_template=False`), and retrieve the same top-12 pack. For the frozen configurations, readout fidelity decreases as the split moves deeper. At the same split $j=12$, the rank-32 LoRA improves the LoCoMo Judge score by 13.75 points and the BABILong qa1, qa2, and qa5 means by 22.2, 9.0, and 8.4 points, respectively. BABILong entries are rounded task means from the underlying 21 task-length cells; the raw-cell $j=0$ macro is 56.14. The $j=0$ configuration performs full-depth recomputation over the retrieved pack, providing a selected-pack reference without lower-layer reuse and thus no depth-based reduction in query-time computation.

A.1 Mechanism ablations

Split depth and self-distillation. The probe values plotted in Figure 2 use five stratified data splits for Qwen and OLMo plus a matched-support Llama SST2 result; the protocol is specified above. Table 8 gives the frozen $j = 0/6/9/12$ quality sweep and same- j distilled control; Table 9 gives separately distilled $j = 6/9/12/18$ deployment points. Table 10 provides a separate same- j RULER control. Finally, Table 11 uses an HCache-style checkpoint path only as an interface-adaptation diagnostic, showing that the readout repair transfers beyond CoMem.

Selection. Table 12 compares oracle, BM25, recency, and reader attention. BM25 is near oracle on the lexical needle tasks and is more reliable than the tested training-free alternatives at long lengths. Variable tracking instead requires iterative expansion along literal variable links; a three-round CPU microbenchmark costs 188.8 ms at 128k, versus 41.0 ms for one-shot BM25, without an additional model forward. This is a low-overhead selector in the tested regime, not a constant-time one: lookup grows approximately linearly with store size (Table 23). A frozen BGE-large-en-v1.5 retriever is also evaluated on a focused cohort: it preserves strong hit-conditional reader accuracy but loses recall as context grows.

Split j	LoRA params	RULER B	Full — deployed (95% CI)	LoCoMo	Read (ms)	Read speedup	Write (ms)
6	72.745M	98.29	0.91 [0.49, 1.39]	40.38	830.3	1.166×	133.4
9	65.470M	97.55	1.65 [1.04, 2.31]	39.02	748.3	1.273×	196.9
12	58.196M	96.07	3.12 [2.36, 3.93]	38.27	664.4	1.403×	—
18	43.647M	55.41	43.79 [41.77, 45.85]	28.65	499.5	1.807×	390.3

Table 9: Distilled depth curve. Each split has a separately trained rank-32 adapter on layers $[j:36)$ under the same data order, seed, steps, optimizer, and KL recipe. RULER uses Cohort B ($n=1,500$ paired examples); gap CIs are paired bootstrap intervals against full replay with that depth’s adapter, and all McNemar tests reject equality ($p \leq 3.05 \times 10^{-5}$). LoCoMo reports the full-set GPT-4o judge point estimate ($n=1,986$). Read/Write use a fixed 16k same-pack input and three process medians; retrieval, I/O, and decode are excluded, and total latency remains decode-dominated. The matched fixed-pack Write value was not retained for the earlier $j=12$ artifact. Adapter spans and parameter counts differ, so this is not a training-compute-matched depth ablation.

Task	Adapter	8k	16k	32k
niah_single_2	off	36	7	22
	on	100	100	100
niah_multikey_1	off	8	3	1
	on	91	95	92

Table 10: Separate controlled RULER self-distillation ablation on Qwen3-8B at the same split $j=12$ ($n=100$ per cell). Each on/off pair uses the same task, length, selector, and cohort, so the difference isolates the adapter. Table 8 provides the broader same- j LoCoMo/BABILong control used in the main argument.

Configuration	LoRA	LoCoMo Judge
HCache-style ($j=12$)	no	13.29
HCache-style ($j=12$)	yes	31.17

Table 11: Interface-adaptation diagnostic on an HCache-style, retrieval-free checkpoint path. The two configurations share the $j=12$ split and evaluation harness, isolating the adapter. The gain shows that aligning intermediate states to native upper-layer readout transfers beyond CoMem; this is not a head-to-head long-term-memory comparison.

Task	Len	BM25	Recency	ReaderAttn	Oracle
niah_single	8k	100	100	100	100
	16k	100	100	100	100
	32k	100	82	73	100
niah_multikey	8k	99	98	97	100
	16k	99	88	90	100
	32k	99	54	60	100
var-track	8k	99.4	99.2	99.8	N/A
	16k	92.6	92.4	92.4	N/A
	32k	32.0	41.2	27.8	N/A

Table 12: Single-pass selector sweep on RULER ($n=100$) without a chat template; each entry is the peak over $k \in \{4, 8, 12, 16, 24\}$. Oracle retrieval keeps both needle tasks at 100, showing that their long-range gap is selection rather than readout loss. The variable-tracking rows belong to this specific selector-sweep configuration and should not be conflated with the focused top-12 flagship in Table 14. Oracle is undefined for a multi-chunk reference chain and is therefore not reported there.

We do not train or tune a retriever, so our results establish lexical and frozen-dense operating points rather than selector optimality.

Write-context factorization. The complete 2×2 control is reported in the main paper (Table 5). It confirms that the position-only intervention does not repair chunk-local states and that context and position interact.

Read interaction and chunk size. Table 15 compares full cross-chunk attention with block-diagonal reuse. Table 16 reports the chunk-size sweep; its entries are point estimates rather than a multi-seed stability ranking.

Task / length	qa1 16k	qa2 16k	LongEval 16k	RULER multikey 16k	LoCoMo
Dense recall@12	.52	.45	.58	.81	.70
Raw-text reader quality	.45	.31	.57	.82	.18

Table 13: **Frozen dense-retriever diagnostic.** BGE-large-en-v1.5 uses CLS pooling, L2 normalization, cosine search, the same 512-token chunks, and a Qwen3-8B $j=0$ reader without LoRA ($n=100$ per cell). Long-context recall can be the bottleneck even when hit-conditional reading remains strong; this is a fixed top-12 diagnostic, not a trained-retriever comparison or the main equal-latency result.

Method	RULER task	8k	16k	32k	64k	128k
CoMem	niah_single_2	100.0	100.0	99.0	99.0	100.0
	var-track	96.6	97.6	98.8	99.0	95.8
StreamingLLM	niah_single	85.0	36.0	20.0	10.0	2.0
	var-track	41.8	2.2	0.6	0.2	0.8

Table 14: Equal-budget, no-chat RULER comparison ($n=100$ /cell). CoMem and StreamingLLM use the same nominal budget and scorer but not paired examples; CoMem retrieves about 6.5k relevant tokens, whereas StreamingLLM keeps recent tokens.

Task	Full attn.	Block-diag.	Δ
RULER niah_single	100 / 100	100 / 96	0 / +4
RULER niah_multikey	96 / 94	60 / 32	+36 / +62
BABILong qa2	36 / 20	17 / 13	+19 / +7
BABILong qa5	78 / 69	78 / 55	0 / +14

Table 15: Full cross-chunk attention versus block-diagonal reuse (8k/16k, no chat). RULER uses $n=50$ and BABILong $n=100$ with identical top-12 retrieval. Joint recomputation matters most for multi-fact tasks.

chunk	read tokens	uncached step (s/token)	8k	16k	32k	64k
128	1,665	0.19	91.0	90.0	81.0	85.0
256	3,329	0.37	80.0	90.0	90.0	94.0
512	6,657	0.72	89.0	95.0	97.0	94.0
1024	13,313	1.60	100.0	89.0	92.0	94.0

Table 16: Chunk-size trade-off on RULER niah_multikey ($n=100$, no chat). Packed-step cost grows approximately linearly with read length; accuracy values are point estimates, not a multi-seed claim.

A.2 Full chat-template-free benchmark comparisons

Tables 17–20 give the full audited comparisons for RULER, LongEval, BABILong, and LongBench. Main rows use the unified no-chat protocol; they include KV-Direct (Qasim et al., 2026), In-LLM (Xiao et al., 2024a), StreamingLLM (Xiao et al., 2024b), and MemoryLLM (Wang et al., 2024). MemoryLLM uses its released Llama-3 checkpoint (Grattafiori et al., 2024), while the marked LLoCO row uses its released native setup (Tan et al., 2024). These are descriptive external references rather than matched rankings. The all-method LoCoMo comparison is Table 22; judge details are in Appendix B.

Method	Task	8k	16k	32k	64k	128k
CoMem + LoRA	single-2	100	100	99	99	100
	multikey	95	94	97	91	93
	var-track	96.6	97.6	98.8	99.0	95.8
CoMem frozen ($j=9$)	single	96	99	99	99	96
	multikey	44	44	59	30	48
	var-track	45	38.8	36.4	31.2	25.8
KV-Direct	single	100	100	100	100	0
	multikey	100	100	98	88	0
	var-track	100	100	99.8	96.2	0
InfLLM	single	100	100	92	61	57
	multikey	100	79	65	45	20
	var-track	100	96.8	90.6	81.8	79.2
StreamingLLM	single	85	36	20	10	2
	multikey	83	34	18	13	4
	var-track	41.8	2.2	0.6	0.2	0.8
<i>External persistent-memory reference (different backbone)</i>						
MemoryLLM [†]	single	29	40	37	21	21
	multikey	28	24	25	12	10
	var-track	0	1.2	0	0	0

Table 17: Full chat-template-free RULER comparison using official `string_match_all` ($n=100$ per cell). KV-Direct places the full context in the read with native RoPE and no YaRN; its 64k/128k cells are unextended stress references rather than a length-extended upper bound. CoMem keeps a bounded retrieved pack. This is **Cohort A**: every method uses `niah_single_2`, `niah_multikey_1`, and `variable_tracking`, with $n=100$ and the official scorer. Its CoMem macro is 97.05. Cohort B instead uses `niah_single_3` to match the YaRN arm (Table 21) and yields 96.07; the cohorts are never pooled in a direct comparison. [†]MemoryLLM is the principal external persistent-memory reference and uses its released Llama-3-8B-chat checkpoint; its different backbone precludes matched attribution.

Method	8k	16k	32k	64k	128k	Mean
CoMem + LoRA	69	75	64	67	70	69.0
CoMem frozen ($j=9$)	8	0	4	0	4	3.2
KV-Direct	100	96	92	38	0	65.2
StreamingLLM	86	34	18	10	6	30.8
InfLLM	60	30	12	4	2	21.6
MemoryLLM [†]	22	22	16	6	2	13.6

Table 18: LongEval register-content accuracy. Unmarked rows use the unified chat-template-free protocol. Means use the common 8k–128k support. KV-Direct uses native RoPE without YaRN, so its 64k/128k cells are unextended stress references and the five-length mean is not a length-extended upper bound. CoMem’s separate six-length headline is 72.83 after including its 4k score of 92. The frozen CoMem control used 48 generation tokens rather than 16; both budgets exceed the single-number answer length. [†]MemoryLLM is the external persistent-memory reference and uses a different chat-tuned backbone, so its row is not a matched comparison.

Method	Task	0k	1k	2k	4k	8k	16k	32k	Mean
CoMem + LoRA	qa1	98	80	68	68	46	17	12	55.6
	qa2	26	44	43	44	23	8	1	27.0
	qa5	68	76	76	75	68	60	58	68.7
CoMem frozen ($j=9$)	qa1	98	59	71	57	7	1	4	42.4
	qa2	53	17	28	26	8	4	1	19.6
	qa5	70	74	65	60	41	39	40	55.6
KV-Direct	qa1	98	84	80	74	80	72	63	78.7
	qa2	58	53	50	46	49	49	37	48.9
	qa5	71	73	62	59	65	42	58	61.4
InfLLM	qa1	97	84	81	72	69	52	34	69.9
	qa2	51	58	41	47	41	39	30	43.9
	qa5	69	68	63	66	73	60	55	64.9
StreamingLLM	qa1	97	83	80	71	23	27	12	56.1
	qa2	49	56	44	49	19	11	4	33.1
	qa5	68	65	68	65	39	47	34	55.1
MemoryLLM [†]	qa1	52	46	35	28	23	17	12	30.4
	qa2	37	29	17	18	21	17	11	21.4
	qa5	50	45	39	35	35	34	29	38.1

Table 19: BABILong accuracy ($n=100$ per task-length cell), evaluated with the official `compare_answers` scorer and `TASK_LABELS`. Within the native context window, full-context KV-Direct is substantially stronger on qa1 and qa2, exposing the compression tax, whereas distilled CoMem achieves the highest qa5 mean, ahead of InfLLM (68.7 vs. 64.9). All methods are evaluated under the unified plain-text prompting protocol, with model-specific chat templating disabled (`chat_template=False`). [†]MemoryLLM is the principal external persistent-memory reference and uses its released Llama-3-8B-chat checkpoint. Its different backbone precludes matched attribution; the official chat template does not improve these diagnostics (Appendix A).

Method	NQA	Qasper	Hotpot	2Wiki	MultiF.	Musique	Macro
CoMem + LoRA	4.12	11.01	11.62	12.83	25.41	7.91	12.15
CoMem frozen ($j=9$)	4.63	11.23	9.41	10.71	22.05	5.72	10.63
KV-Direct	3.70	11.82	12.68	12.03	25.30	7.49	12.17
InfLLM	2.99	11.35	12.08	12.50	25.33	6.94	11.86
StreamingLLM	3.49	11.64	11.02	12.19	22.42	5.88	11.11
MemoryLLM [†]	3.13	8.46	8.76	10.27	17.43	5.98	9.01
LLoCO [‡]	24.21	24.45	44.24	—	—	—	30.97

Table 20: Official LongBench QA token F1 by dataset. Except for the explicitly marked LLoCO diagnostic, methods use the same plain-text no-chat protocol. CoMem with LoRA scores 12.15 and full-context KV-Direct scores 12.17; because only CoMem uses the split-interface adapter, this near-equality is descriptive rather than a training-budget-matched compression estimate. The frozen CoMem configuration reports adapter-free performance at $j=9$. NQA denotes NarrativeQA, and MultiF. denotes MultiFieldQA-en. [†]MemoryLLM is the principal external persistent-memory reference and uses its released Llama-3-8B-chat checkpoint. Its different backbone precludes matched attribution; its official chat template does not improve the tested diagnostics (Appendix A). [‡]LLoCO uses LLaMA-2-7B AutoCompressor, released per-domain supervised LoRAs, fp16, and its native task prompts. Released weights cover only NarrativeQA/Qasper/HotpotQA on our 200-example sets; 30.97 is their three-task macro and is not comparable to the six-task macro column.

Task	Method	8k	16k	32k	64k	128k
single needle	KV-Direct	100	100	100	100	0
	KV-Direct+YaRN [†]	100	100	100	100	100
	CoMem+LoRA	100	91	97	98	98
	CoMem+LoRA+YaRN [†]	100	90	97	97	98
multikey	KV-Direct	100	100	98	88	0
	KV-Direct+YaRN [†]	98	94	96	91	89
	CoMem+LoRA	94	91	99	90	93
	CoMem+LoRA+YaRN [†]	82	85	94	88	93
variable tracking	KV-Direct	100	100	99.8	96.2	0
	KV-Direct+YaRN [†]	99.2	99.4	26.6	67.2	57.8
	CoMem+LoRA	96.2	98.0	98.2	98.6	99.0
	CoMem+LoRA+YaRN [†]	88.6	90.0	89.4	95.0	93.6

Table 21: Matched- $n=100$ Qwen3-8B RULER scaling. [†]YaRN factor 4 gives an effective 163,840-token window. At 128k, unextended-backbone CoMem+LoRA scores 98/93/99.0 on single-needle, multikey, and variable tracking, versus 100/89/57.8 for KV-Direct+YaRN. Applying YaRN to CoMem changes these to 98/93/93.6, showing an in-window position-scaling tax on variable tracking.

Method	Judge	Judge _{1:4}	F1	Substr. acc.
CoMem + LoRA ($j=12$)	38.27	48.64	9.15	23.36
KV-Direct	34.59	43.83	9.02	22.36
StreamingLLM	25.63	31.04	7.67	13.75
InfLLM	22.21	26.43	7.39	13.34
MemoryLLM [†]	16.11	20.65	5.91	9.52

Table 22: LoCoMo comparison on all 1,986 chat-template-free examples. Judge_{1:4} is GPT-4o accuracy on the 1,540 answerable items; the full Judge column folds in local abstention scoring for 446 adversarial items. F1 and substring accuracy are deterministic lexical diagnostics; semantic Judge is the primary metric. All rows use the same scoring protocol. The same- j adapter effect is isolated in Table 8. On paired answerable items, distilled CoMem exceeds KV-Direct by 4.81 points (item-bootstrap 95% CI [2.34, 7.34]); a 10-conversation cluster bootstrap gives [1.27, 8.32] and 8/10 observed conversation differences favor CoMem. An independent DeepSeek-V3 audit preserves the ordering. [†]MemoryLLM is the external persistent-memory reference and uses its official chat-tuned Llama-3 checkpoint; its different backbone precludes matched attribution. Its native chat template does not improve this result.

A.3 Store scaling and failure boundaries

Table 23 tests the “bounded read over extensible store” claim directly. Retrieval is invariant when the required evidence fits top-12, but lookup and index construction still scale with store size. Beyond the budget, recall is physically capped; co-keyed multivalued evidence falls below even this cap (recall 0.62 at $E=8$), and a common-word-frequency task scores zero as coverage falls from 4.5% at 128k to 1.2% at 512k. These negative results delimit the claim: bounded retrieval is not global aggregation, and successful retrieval does not guarantee multi-hop reasoning.

We audited the PG-19 adapter corpus against the InfiniteBench long-book support using normalized 13-gram containment sketches. The conservative 0.8 threshold flags 24/86 books (163/580 records), but containment is sharply bimodal: 31 books are below 0.10, no book lies in 0.18–0.60, and 54 are at least 0.60. Any cut in 0.20–0.60 therefore flags 54/86 books and 387/580 records; anonymized character names make the 0.8 count an undercount. Existing predictions were unavailable for a strict-clean-subset rescore (113 QA and 76 choice records), so we remove the book-quality comparison rather than treat it as out-of-domain evidence. This does not affect the synthetic store-scaling results above. We did not complete equivalent overlap audits for every natural benchmark, including NarrativeQA; those results should therefore be interpreted as benchmark performance rather than contamination-free generalization estimates.

A.4 Reproducibility details

Tables 24–26 record the configuration, training, and evaluation settings used by the Qwen3-8B flagship. The archived configuration files and evaluation scripts accompany the code release.

Artifacts, licenses, and data scope. The study evaluates English text only. It uses the frozen released Qwen/Qwen3-8B checkpoint (Apache-2.0); we do no backbone continued pretraining. It also uses PG-19 (Rae et al., 2019), RULER (Hsieh et al., 2024), BABILong (Kurato et al., 2024), LongBench (Bai et al., 2023), LongEval/LongChat (Li et al., 2023), and LoCoMo (Maharana et al., 2024). RULER and LongChat code are Apache-2.0; the LongBench repository is MIT-licensed, while its component datasets retain their upstream terms; BABI-

Long code is Apache-2.0 and incorporates BSD-licensed bAbI tasks; LoCoMo is CC BY-NC 4.0. PG-19 contains pre-1919 Project Gutenberg books, for which the dataset page does not assert one uniform corpus license, so we do not redistribute its text. We likewise redistribute no model weights, benchmark examples, or private API responses. Subject to upstream terms, the anonymous archive includes the adapter, source, documentation, pinned requirements, prediction hashes or permissible shards, judge prompts/parsed outputs, raw timing records, and third-party notices. Users obtain all external artifacts from their original providers and remain responsible for their terms. Table 26 reports the evaluated supports and sample counts; LoCoMo category denominators are in Appendix B.

Software environment. The archived environment specifies Python 3.10+, PyTorch 2.10, Transformers 5.5.4, PEFT 0.10+, Datasets 2.14+, bf16, and SDPA. Evaluation uses the benchmark-specific official scorers named in Table 26; the archive records all non-default flags and provides a CPU correctness gate.

Compute accounting. The final adapter run used one node with 8 H20 GPUs for about 22 minutes, or approximately 2.9 H20 GPU-hours. Table 25 and this section report model size, trainable parameters, hardware, and measured timing protocol. Total GPU-hours across preliminary probes, failed runs, baseline generation, and all ablations were not consistently logged; we report this omission rather than extrapolating an unreliable total.

Efficiency measurement. The Write-inclusive cohort uses one NVIDIA L20A with PyTorch 2.10, Transformers 5.5.4, bf16, SDPA, and the rank-32 LoRA enabled, reporting medians of three repetitions. Full-context prefill times the complete forward; CoMem times document Write, CPU retrieval, query Write, and upper-layer Read through logits. The timer excludes one-time BM25 index construction and external-store I/O; Table 30 measures the latter separately. At 128k, full-context prefill is 6.035 s and peaks at 89.39 GB, while CoMem takes 2.202 s and peaks at 18.54 GB. Cached decode uses 20 greedy steps.

An earlier representative operating-point cohort is a separate one-H20 bf16/SDPA sweep; its A/B quality values are not a single matched Pareto frontier. Retrieved rows share top-12 IDs, a 6,657-

Store	NIAH recall	NIAH score	VT recall	VT score	Read tokens	BM25 (ms)
128k	1.00	100	1.00	18	6,211 / 6,463	80 / 20
256k	1.00	100	1.00	34	6,209 / 6,463	164 / 39
512k	1.00	100	1.00	40	6,209 / 6,464	328 / 81
1M	1.00	100	1.00	54	6,209 / 6,463	697 / 170
2M	1.00	90	1.00	38	6,211 / 6,463	1,407 / 357
4M	1.00	100	1.00	34	6,208 / 6,463	2,852 / 725

Table 23: Store scaling with a fixed top-12 read. Values separated by “/” are single-evidence NIAH / five-link variable tracking (VT). Evidence recall ($n=20$) stays perfect and the model read stays about 6.2–6.5k tokens from 128k to 4M stored tokens; BM25 lookup grows approximately linearly. Generation scores use $n=10$ and reveal that VT reasoning remains noisy even when all five evidence chunks are retrieved. At 128k, increasing required VT evidence from 12 to 16/24/32 gives recall 1.0 to 0.75/0.50/0.375, exactly $\min(1, 12/E)$.

Layer	Setting	Value
Backbone	Architecture	Frozen Qwen/Qwen3-8B; revision b968826d9c46 (full hash in artifact manifest); $L=36$, $d=4096$, 32 query heads, 8 KV heads, bf16, SDPA.
Backbone	Positions	Native window 40,960; RoPE $\theta=10^6$; no active RoPE scaling or YaRN.
Write	Cached state	Split $j=12$; disjoint 512-token chunks; chunk-local positions restart at zero; one bf16 residual tensor h_j per token.
Read	Pack	[BOS; selected h_j ; $h_j(q)$] in document order, with fresh contiguous RoPE and full causal cross-chunk attention. The nominal cap is $1 + 12 \times 512 + 512 = 6,657$ tokens; actual RULER means are about 6.2–6.5k because tail chunks and queries can be shorter.
Retrieval	BM25	Token-ID documents and bare-question token IDs; $k_1=1.5$, $b=0.75$, Robertson IDF with $+1$ smoothing; no lowercasing, stemming, or stop-word removal.
Retrieval	Iteration	$\text{rounds}=0$ means automatic $\lceil k/h \rceil$ rounds, not one-shot. Each round uses newly selected chunks as the next frontier and deduplicates the selected set.
Retrieval	Width	All five benchmarks fix $k=12$, hop width $h=4$, and therefore three automatic rounds. Chunk size 512 and the BOS sink are also fixed.
Read	Mask	The pack is one causal sequence: the final query sees all chunks; later chunks may attend to earlier chunks. In the block-diagonal ablation, chunks see only themselves and the sink, while the final query still sees all chunks.
Generation	Decoding	Greedy decoding (no sampling, temperature, top- p , or beams); no chat template and no thinking mode.
Adapter-free arm	Deployment	Frozen backbone, no LoRA, $j=9$; all other memory and evaluation settings are held fixed.

Table 24: Flagship CoMem inference configuration. The retrieval and read configuration is frozen across all five benchmarks; only benchmark harness parameters differ.

Setting	Value
Adapter	Rank 32, $\alpha=64$, dropout 0; Q/K/V/O and gate/up/down projections in layers 12–35 only; 58.20M trainable parameters (0.71%). Final weight SHA-256: dd09cd17457c63578c0f38dab79b287ab5da6e3f14c119aedefec1c34400536f.
Teacher and objective	The same Qwen3-8B with adapters disabled and $j=0$ serves as the frozen teacher. On the teacher top-64 support, $\mathcal{L} = 0.6 \text{KL}(p q) + 0.4 \text{KL}(q p)$ with implicit temperature 1 and no CE term. The loss is applied only to the query segment.
Data	Plain PG19 train text; no synthetic long-context examples, retrieval, task labels, or benchmark supervision. Training uses 2,048-token windows, formed as four 512-token chunks.
Optimization	AdamW, $\beta = (0.9, 0.95)$, weight decay 0, peak LR 8×10^{-5} , 100-step linear warmup, cosine decay to zero, gradient clipping 1.0, seed 42, bf16, no gradient checkpointing.
Schedule	4,000 steps on 8 GPUs, one sample per GPU and no gradient accumulation (global batch 8), totaling about 65.5M tokens.
Compute	One node with 8×NVIDIA H20 GPUs. Logged throughput is about 24.5 samples/s, implying roughly 22 minutes wall-clock; training peak memory was not recorded.

Table 25: Self-distillation and LoRA training configuration. The adapter name’s “4k” denotes 4,000 optimization steps; the training sequence length is 2,048 tokens.

token nominal pack (measured ~ 6.5 k), and chunk size 512. Write-all timing processes the complete store once; Read is per query. The $j=0$ text-retrieval path has no adapter, whereas $j=12$ uses the flagship LoRA; hence the 0.33 s Read difference is an operating-point comparison rather than a depth-only effect. Its 5.74 s extra Write implies a 17–20-query analytic break-even in that earlier component cohort. The main serving experiment instead measures write-once/fetch/reuse directly and obtains about 8–11 queries at 32k and 26–28

at 128k; the difference comes from source length, generation length, storage tier, and timing boundary. CoMem stores 8 KB/token (about 1 GiB at 128k), far more than text but $18\times$ less than full KV. Main timing cohorts keep the store in HBM; Table 30 separately measures placement beyond HBM, so its I/O microbenchmark is not added to model latency.

Retained-KV references. Table 27 reports SnapKV and PyramidKV (Li et al., 2024; Cai

Benchmark	Tasks and support	Generation	Scoring and aggregation
RULER	Single needle, multikey, and variable tracking; 8k–128k; 100 examples/cell; 8 shards; seed 42.	48 tokens; 60 for variable tracking.	Official case-insensitive <code>string_match_all</code> ; equal-weight macro over 15 cells.
LongEval	Lines retrieval; 4k–128k; 100 examples/length; 8 shards; seed 1234.	16 tokens for the flagship; 48 for frozen controls.	First numeric answer exact match; six-length macro (five-length 8k–128k macro for matched baseline tables).
LongBench	Six QA datasets; 1,150 examples total; 8 shards; seed 42.	32/64/128 tokens by dataset.	Official multi-reference token F1; equal-weight macro over six datasets.
BABILong	qa1/qa2/qa5; 0k–32k; 100 examples/cell; 4 shards; seed 42.	20 tokens.	Official <code>compare_answers</code> with <code>TASK_LABELS</code> .
LoCoMo	All 1,986 QA pairs from 10 conversations; 4 shards.	48 tokens for the main 8B comparison; 512 for the cross-scale audit.	GPT-4o judge on categories 1–4; official local abstention rule for category 5; full-set judge is the headline.

Table 26: Benchmark-level evaluation settings. Every main comparison uses `chat_template=False`. MemoryLLM is the external persistent-memory reference, evaluated with its released chat-tuned backbone.

et al., 2024) under their full-prefill boundary. They answer a different systems question from cross-query persistence and are not treated as nearest modular-cache baselines.

Paired quality–latency replay control. Table 28 uses a separate H20 harness. The two arms share the Qwen3-8B checkpoint, flagship LoRA SHA-256 and all 168 mounted modules, example IDs, decode parameters, and per-example retrieved chunk IDs, order, and pack tokens; only replay start differs. Across 15 RULER Cohort-B cells ($n=100$ each), full replay scores 99.19 and layer-12 replay scores 96.07. The paired 3.12-point gap has bootstrap 95% CI [2.36, 3.93] and exact McNemar $p=8.79 \times 10^{-24}$ (83 full-only correct versus one layer-12-only correct). The gap is largest on distractor-heavy multikey retrieval.

For latency, each arm uses the same approximately 6.5k-token pack retrieved from 16k source contexts in three independent processes, with three warmups and 20 reads per process. Layer-12 replay is $1.403\times$ faster (931.9 to 664.4 ms), with all three process-level ratios between 1.402 and 1.404. Total-decode medians remain similar at roughly 2.76–2.86 s. A preceding component harness attributes the read reduction primarily to upper-transformer forward ($1.398\times$), with unchanged LM-head time. Retrieval, index construction, reusable store/query Write, and persistent I/O are outside this read-phase timing. The result is therefore a measured quality–latency trade-off, not quality-preserving or end-to-end acceleration.

A.5 Training-seed robustness

We trained two additional $j=12$ rank-32 LoRA adapters on the same data budget and evaluated all three seeds, including the flagship, on identical cells and sample counts (RULER $n=50$, BA-

BILong $n=100$). Across the 18 reported RULER and BABILong cells, the median cell-wise standard deviation is 1.34 points and the maximum is 4.36; the flagship falls within the observed range on the RULER headline cells and is, if anything, slightly conservative on the two BABILong cells that drive the maximum. The two added seeds use effective batch 3 rather than 8, so this remains a matched-data-budget robustness check with slightly different optimization noise rather than a fully controlled three-seed estimate.

A.6 Prompt and native-chat sensitivity

For Qwen3-8B, the GPT-4o LoCoMo judge is substantially less prompt-sensitive than token F1. CoMem scores 38.27 without chat templating and 37.76 with it; KV-Direct scores 34.59 and 38.22. In contrast, CoMem’s F1/substring accuracy changes from 9.15/23.36 to 19.51/28.65, while KV-Direct F1 changes from 9.02 to 40.06, showing that lexical metrics are strongly formatting-sensitive. We therefore retain the uniform no-chat judge comparison as the controlled headline.

The official MemoryLLM chat template does not improve the tested BABILong, RULER, or LoCoMo diagnostics; full results are included in the released artifacts. We therefore keep the uniform no-chat row as the external persistent-memory reference and do not mix prompt protocols within an average.

A.7 Cross-scale and MoE generality

Adapter-free sweeps across six Qwen3 sizes and Qwen3-30B-A3B confirm that the split-forward implementation ports beyond the flagship. Performance is non-monotonic because split, architecture, and task differ, so we infer neither a scaling law nor a replicated quality frontier. These exploratory values and raw predictions are re-

(a) Quality at a 6,657-token/KV budget					
Method	Native A macro	Full-prompt single	64k/128k multikey	tracking	LoCoMo F1/acc
CoMem+LoRA	97.05	99/100	91/93	99.0/95.8	9.15/23.36
SnapKV	72.33 [†]	100/100	94/91	80.0/84.8	9.21/22.05
PyramidKV	72.32 [†]	100/100	93/89	88.6/86.8	9.13/22.10

(b) Full-prefill systems cost on one H20				
Method	Len.	ms	GB	ms/tok
SnapKV	8k	1,213	18.6	24.7
	32k	6,342	25.7	24.6
	128k	51,520	54.3	24.7
PyramidKV	8k	1,203	18.6	24.7
	32k	6,301	25.7	26.7
	128k	51,520	54.3	26.8

Table 27: Standard selected/compressed-KV baselines at CoMem’s retained budget. Panel (a) uses RULER Cohort A and full-set LoCoMo lexical scoring. CoMem reads its bounded pack from the full store without YaRN; the SnapKV/PyramidKV 64k/128k columns use YaRN factor 4 and see the full prompt. [†]Their native 15-cell macros include 64k/128k inputs left-truncated to the 40,960-token native window; within 8k–32k both score 99.96. Panel (b) times full-prompt prefill, compression, and 64-token decode (median of three); the final column is decode latency. These methods must prefill the full prompt before eviction. CoMem’s separate L20A cohort peaks at 18.54 GB at 128k (Appendix A.4); hardware differs, so we report no cross-table latency ratio.

Replay start	RULER	Read (ms)	p10	p90
$j=0$	99.19	931.9	931.6	942.0
$j=12$	96.07	664.4	663.8	667.1
Trade-off	−3.12 pt	1.403× speedup		

Table 28: Paired quality–latency control with the same LoRA. Quality is the 15-cell RULER macro ($n=1,500$ paired examples); the 3.12-point loss has paired-bootstrap 95% CI [2.36, 3.93]. Latency uses a fixed approximately 6.5k-token pack retrieved from 16k source contexts and reports medians across three independent processes (20 reads each). Retrieval, reusable Write, I/O, and total decode are excluded; total decode is similar across arms.

Replay pathway	RULER B	MK 8/16k	Reusable
Full replay from $j=0$	99.19	100.0	No
Continuous-prefix h_{12} oracle	99.19	100.0	No
Chunk-local cached h_{12}	96.07	92.5	Yes

Table 29: Write-side attribution with identical examples, packs, and LoRA. The continuous-prefix oracle runs layers 0–11 over the selected packed sequence for every query, then starts upper replay from that h_{12} ; it is bit-identical to full replay on all 1,500 Cohort-B examples. The deployable chunk-local path loses 3.12 points (95% CI [2.36, 3.93]; McNemar $p=8.79\times 10^{-24}$), and 7.50 points on the pooled 8k/16k multikey cells ($n=200$, CI [4.0, 11.5]). The oracle is an attribution upper bound, not a reusable cache.

Backend	Write GB/s	Fetch ms	H2D ms	Peak QPS	16M outcome
GPU resident	OOM	–	–	–	128 GiB > 95 GiB
CPU pinned	139.6	0.81	1.41	941	235 GB host
Local NVMe	2.45	13.42	0.91	264	2.18 GB host
Network/CEPH	1.66	79.20	1.20	42	2.19 GB host

Table 30: Measured persistent-store tiers at 16M tokens (128 GiB of bf16 h_{12}). Each query fetches the same 50.3 MB top-12 pack; medians use seven repetitions after two warmups, and QPS is the best of 1/4/16 concurrent clients. GPU residence fits 8M tokens (64.98 GB peak) but not 16M; off-GPU tiers retain store-size-independent transfer volume at increasing I/O latency. This microbenchmark times storage and transfer, not BM25 or model inference.

tained in the artifact; the Hy3 study below gives the stronger architecture-level partition check reported in this appendix.

A.7.1 Hunyuan Hy3: an 80-layer sparse MoE

We additionally port CoMem to Tencent Hunyuan Hy3 (Tencent Hunyuan Team, 2026), an 80-layer sparse Mixture-of-Experts model (192 experts, top-8 routing plus one shared expert; ≈ 597 GB in bf16) with a 262,144-token native window. The backbone is sharded across eight GPUs; Write and Read semantics are unchanged.

Exact partition through MoE routing. Reconstructing a contiguous forward by writing `layers[0:j]` and resuming `layers[j:80]` gives $\max|\Delta\logit| = 0.0$ at $j \in \{0, 1, 40, 80\}$, including the Dense-to-MoE boundary. The resumed half re-executes every router and expert, so the partition preserves routing exactly on this self-test.

Split depth and self-distillation. A PG19 sweep places the best readout region near $j \approx 32$ –36. At $j = 32$, self-distillation reduces the 16k

multiplicative LM tax from 1.496 to 1.078 and raises top-1 agreement from 0.773 to 0.871 (Table 31).

$j=32$	ctx 8k		ctx 16k	
	gap	top1	gap	top1
zero-shot	1.378	0.823	1.496	0.773
distilled	1.071	0.895	1.078	0.871

Table 31: CoMem self-distillation on Hy3 at split $j=32$ (LoRA $r=32, \alpha=64$ on `layers[32:]`, teacher $j=0$ RAG recompute, 4000 steps of plain PG19 KL, no synthetic data). **gap** is the multiplicative LM tax (CoMem-readout perplexity / full-context perplexity; 1.0 is exact); **top1** is argmax agreement with the full forward, over matched 16-doc PG19 windows. The 16k LM tax falls from +49.6% to +7.8% and top1 rises $0.773 \rightarrow 0.871$, reproducing the 8B-backbone self-distillation effect on the 80-layer MoE.

Bounded read through 256k. With the distilled adapter and BM25 top-8, the read stays at ≈ 4.3 –4.6k tokens from 16k through 256k. CoMem reaches 100/98 single/multikey recall at the 256k native ceiling with no OOM (Table 32).

Task	16k	32k	64k	128k	256k
niah_single_2	98	92	100	98	100
niah_multikey_1	100	94	100	100	98
read length (tok)	4.3k	4.6k	4.5k	4.4k	4.5k
context (tok)	16k	32k	65k	131k	261k

Table 32: CoMem on the Tencent Hunyuan Hy3 backbone (hy_v3, 80-layer sparse MoE), RULER needle recall ($n=50$, official `string_match_all`), split $j=32$, BM25 top-8, self-distilled LoRA. The read length fed to the model stays ≈ 4.3 –4.6k tokens while the context grows $16\times$ ($16k \rightarrow 256k$); CoMem retains 100/98 recall at the backbone’s 256k native ceiling, with 0 out-of-memory failures. The context row reports the mean packed length of the RULER samples.

B Chat-Template-Free Statistical Verification

All statistics in this appendix were recomputed from the saved prediction shards on the shared result filesystem, without GPU reruns. The Qwen3-8B flagship uses the self-distilled $j=12$ adapter, chunk 512, top-12 retrieval, a BOS sink, and no chat template. The released adapter artifact is identified by the SHA-256 and configuration in Table 25. The main RULER directory uses the

Task	8k	16k	32k	64k	128k
niah_single_3	100.0	91.0	97.0	98.0	98.0
niah_multikey_1	94.0	91.0	99.0	90.0	93.0
variable_tracking	96.2	98.0	98.2	98.6	99.0

Table 33: RULER Cohort B, used for the matched length-extension and $j=0$ comparisons. The $n=100$ macro is $1441.0/15 = 96.0667$; the exact single-needle variant is `niah_single_3`.

`iter_bm25` implementation; the saved configuration records `rounds=0` and an observed read of approximately 6.5k tokens. Every reported RULER cell contains eight of eight shards, 100 examples, no empty prediction, and zero mismatch between the stored score and a fresh official-kernel recomputation. BABILong likewise contains 100 valid examples in every cell.

B.1 RULER cohort definitions

Cohort A is the all-method table (Appendix Table 17): it substitutes `niah_single_2`, keeps the same multikey/tracking tasks and $n=100$, and gives CoMem 97.05. Cohort B above was freshly sampled with `niah_single_3` because that is the exact task available for both YaRN arms and $j=0$; it gives 96.07. Tables mark every RULER aggregate with *A* or *B*, never subtract across them, and exclude legacy 256k cells. Table 21 contains only Cohort B.

B.2 LoCoMo judge protocol and denominator

Prediction shards for all methods and the adapter-free CoMem control deduplicate to all 1,986 LoCoMo items. GPT-4o judges the 1,540 answerable items in categories 1–4. Category 5 contains 446 adversarial abstention items and is scored locally: an answer is correct iff it is empty or matches the refusal rule. Thus the canonical full-set judge is over all 1,986 items, not only the API-judged subset.

Method	Cat. 1	Cat. 2	Cat. 3	Cat. 4	Cat. 5	Overall
CoMem + LoRA	26.95	19.00	30.21	69.32	2.47	38.27
CoMem frozen ($j=9$)	18.79	13.40	27.08	53.75	1.12	29.15
KV-Direct	24.11	18.69	25.00	62.19	2.69	34.59
StreamingLLM	22.70	11.21	23.96	42.21	6.95	25.63
InfLLM	18.44	14.33	25.00	33.89	7.62	22.21
MemoryLLM	15.60	8.72	15.63	27.47	0.45	16.11

Table 34: LoCoMo GPT-4o judge by category. Categories 1–4 contain 282/321/96/841 judged items; Category 5 contains 446 locally scored abstention items. Overall aggregates all 1,986 examples.

The judge uses an OpenAI-compatible chat-completions endpoint with model name `gpt-4o`, `seed=1`, and no client-set temperature or top- p . The endpoint does not expose a dated model snapshot. Requests are retried four times with exponential backoff; unparsable responses and API failures are conservatively scored wrong. The prompt presents the question, gold answer, and prediction; it accepts any semantic match (including equivalent date phrasing), rejects contradictions, omissions, empty answers, and refusals, and requires exactly `CORRECT` or `WRONG`. The verbatim template is released with the evaluation script. Gold alternatives are joined with “OR.” A reply must begin with `CORRECT` or `WRONG`; a fallback substring vote is used before defaulting to wrong.

On the common GPT-4o-judged items ($n=1540$), CoMem scores 48.64 and KV-Direct 43.83. A paired per-item bootstrap of their difference (10,000 resamples, seed 1234) gives +4.81 points with 95% CI [2.34, 7.34]. Because the questions are nested within only 10 conversations, this interval is not our sole inferential check. A paired conversation-cluster bootstrap, resampling the 10 conversations rather than individual questions (same 10,000 resamples, seed 1234), gives the same +4.81 point estimate with 95% CI [1.27, 8.32] — wider than the per-item interval, as expected with only ten clusters; 8 of the 10 observed conversation-level differences favor CoMem. We do not interpret the bootstrap tail fraction as a calibrated hypothesis-test p value. We report the latter as a dependence-aware robustness check rather than claiming that 1,540 questions are independent. The canonical full-set point scores are 38.27 and 34.59, respectively.

B.3 Independent-judge audit

We reran the binary prompt with DeepSeek-V3 on a stratified 200-item subset shared by CoMem and KV-Direct. Agreement with GPT-4o is 0.81 ($\kappa = 0.626$). Both judges preserve the ordering: GPT-4o scores 36.5/32.5 (+4.0) and DeepSeek-V3 56.0/49.0 (+7.0) for CoMem/KV-Direct. Absolute calibration differs, but the positive gap is judge-robust.

B.4 Uncertainty and aggregation checks

Per-method, 1,000-resample item-bootstrap intervals (seed 1234) are [36.20, 40.58] for CoMem’s full-set LoCoMo judge, [32.48, 36.71] for KV-Direct’s, [46.04, 50.98] and [41.23, 46.30] for their

category-1–4 judge scores, and [8.60, 9.72] and [8.51, 9.61] for their F1 scores. These unpaired, item-level intervals are descriptive; the paired cluster analysis above is the appropriate dependence-aware comparison.

We do not reuse cell-bootstrap intervals from the superseded RULER cohort or legacy BAILong predictions. The separate six-length LongEval aggregate is 72.83 over 600 examples (95% CI [69.33, 76.33]); the benchmark summary uses the predefined five-length 8k–128k mean of 69.0.